# Unit-1: Introduction

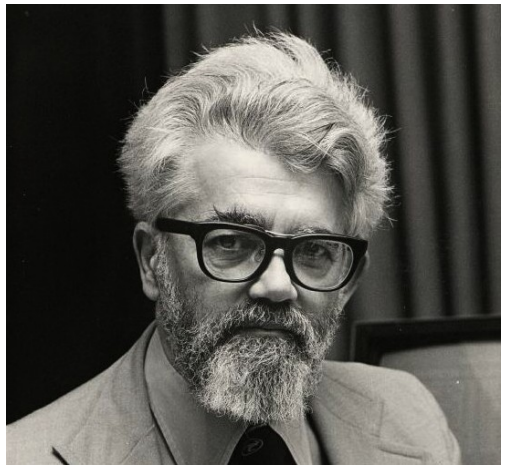


# Outline

- Introduction to Artificial Intelligence (AI)
- Applications of AI

# Introduction to AI

# What is Artificial Intelligence (AI)?



**John McCarthy**
(the father of Artificial Intelligence)

*AI is the science and engineering of making intelligent machines, especially intelligent computer programs (1956).*

- AI is a branch of computer science dealing with the simulation of intelligent behavior in computers.
- AI is the study of how to make computers do things which, at the moment, people do better.
- AI is, the study and design of intelligent agents where an intelligent agent is a system that perceives its environment and takes actions.

# Artificial intelligence – Four Category

| **Thinking Humanly** | **Thinking Rationally** |
| --- | --- |
| "The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985)<br><br>"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978) | "The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985)<br><br>"The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992) |
| **Acting Humanly** | **Acting Rationally** |
| "The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)<br><br>"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991) | "Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998)<br><br>"AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998) |

# Artificial intelligence – Four Category

| Aspect | Thinking Humanly (Cognitive Modeling Approach) | Thinking Rationally (Laws of Thought Approach) | Acting Humanly (Turing Test Approach) | Acting Rationally (Rational Agent Approach) |
|---|---|---|---|---|
| **Definition** | Focuses on making machines **think like humans** by imitating human mental processes. | Focuses on making machines **reason correctly using logic** and formal rules. | Focuses on making machines **behave like humans**. | Focuses on making machines **act optimally to achieve goals**. |
| **Primary Goal** | Replicate human thinking. | Achieve correct logical reasoning. | Produce human-like behavior. | Achieve the best possible outcome through rational actions. |
| **Key Question** | *How do humans think?* | *How should reasoning be done correctly?* | *Can a machine behave like a human?* | *Can a machine choose the best action?* |
| **Basis** | Human psychology and cognitive science. | Logic and formal reasoning. | Human behavior and interaction. | Rational decision-making and intelligent agents. |
| **Main Idea** | Understand how humans learn, remember, solve problems, and make decisions, then build systems that imitate those processes. | Gather facts, apply logical rules, and derive correct conclusions. | If people cannot distinguish a machine from a human, it is considered intelligent. | An intelligent system perceives its environment and chooses actions that maximize success. |
| **Examples** | Human Memory Models, Human Learning Simulations | Expert Systems, Logic Programs, Theorem Provers | Chatbots, Virtual Assistants, Conversational Agents | Self-Driving Cars, Robots, Recommendation Systems, Autonomous Drones, Chess Programs |
| **Applications** | Cognitive Science, Psychology Research, Human Learning Models | Expert Systems, Automated Reasoning, Mathematical Proof Systems | Customer Support Bots, Virtual Assistants, Human-Computer Interaction | Robotics, Autonomous Systems, Reinforcement Learning, Modern AI Systems |
| **Advantages** | • Helps understand human intelligence.<br>• Useful in psychology and cognitive science. | • Produces logically correct conclusions.<br>• Easy to verify reasoning. | • Easy benchmark through Turing Test<br>• Focuses on practical performance. | • Works well in real-world environments.<br>• Handles uncertainty.<br>• Focuses on achieving goals efficiently. |
| **Limitations** | • Human thinking is not fully understood.• Humans can be irrational. | • Real-world problems involve uncertainty.<br>• Human reasoning is not always logical. | • Can appear intelligent without real understanding.<br>• Human imitation is not always optimal. | • Determining the "best" action can be difficult.<br>• Requires accurate information and models. |

# Artificial intelligence – Four Category

1. Thinking Humanly (Cognitive Modeling Approach)

"What happens in your brain when I ask 15 × 18?“

You may answer:

recall multiplication

break it into smaller parts

calculate mentally

You follow a thinking process.

**Definition**

A machine is intelligent if it **thinks like a human being**.

# Artificial intelligence – Four Category

2. Acting Humanly (Turing Test Approach)

**Examples**

ChatGPT

Voice assistants

**Definition**

A machine is intelligent if it **behaves like a human**.

It does not matter how it thinks.

Only the behavior matters.

# Artificial intelligence – Four Category

3. Thinking Rationally (Laws of Thought)

**Example**

Solve this puzzle.

All humans are mortal.

Socrates is a human.

"Socrates is mortal.“

This is logical reasoning.

# Artificial intelligence – Four Category

3. Thinking Rationally (Laws of Thought)

**Examples**

Expert Systems

Rule-based AI

Logic Programming

Prolog

**Definition**

The machine should think **logically**.

Every decision must follow logical reasoning.

# Artificial intelligence – Four Category

4. Acting Rationally (Rational Agent)

**Example**

Google Maps

Suppose there are two routes.

Route A: 25 minutes

Route B: 40 minutes

Google chooses Route A.

Did it think like a human? No.

Did it behave like a human? Not necessarily.

It simply selected the best action.

That is a Rational Agent.

# Artificial intelligence – Four Category

4. Acting Rationally (Rational Agent)

**More examples**

Self-driving cars

Chess AI

Recommendation systems

Robots

Medical diagnosis AI

**Definition**

The machine chooses the **best possible action** to achieve a goal.

It does not have to think like humans.

It simply needs to produce the best outcome.

# Artificial intelligence – Four Category

**Summary:**

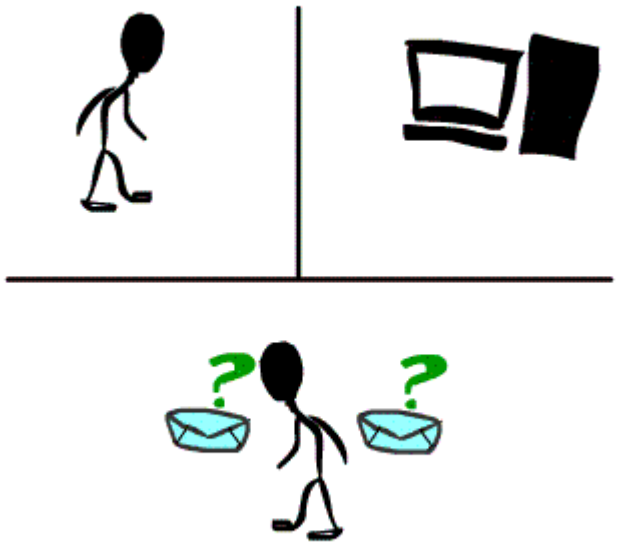
**Thinking Humanly** → Human brain

**Acting Humanly** → Human behavior

**Thinking Rationally** → Logic

**Acting Rationally** → Best action

# Turing Test



- Acting Humanly: The Turing Test proposed by Alan Turing (1950)
- A Turing Test is a method of inquiry for determining whether or not a computer is capable of thinking like a human being.
- The interrogator job is to try and figure out which one is human and which one is computer by asking questions to both of them.
- The computer would try to remain indistinguishable from human as much as possible

# AI Techniques

- There are three important AI techniques:

- Search –
  - Provides a way of solving problems for which no direct approach is available.
  - It also provides a framework into which any direct techniques that are available can be embedded.
  - Search is used when the solution is **not directly known**.
  - Eg: From Home to MSU

- **AI Examples:**

Google Maps finding shortest route

Chess playing (finding best move)

Solving puzzles like Sudoku

Robot path planning

Game playing

Route optimization

# AI Techniques

- Use of knowledge –
  - Provides a way of solving complex problems by exploiting the structure of the objects that are involved.
  - Knowledge allows AI to solve problems by using previously learned facts and rules instead of searching everything from scratch.
  - Domain Knowledge (Rules , Facts, Relationships)

## AI Examples

Medical diagnosis

ChatGPT

Expert systems

# AI Techniques

- Abstraction – Abstraction means ignoring unnecessary details and focusing only on the important information needed to solve a problem.

  ➥ Image Processing

- Face : 🙂

- **Is this real face?**
- **No.**

- Important feature: Eyes, Nose

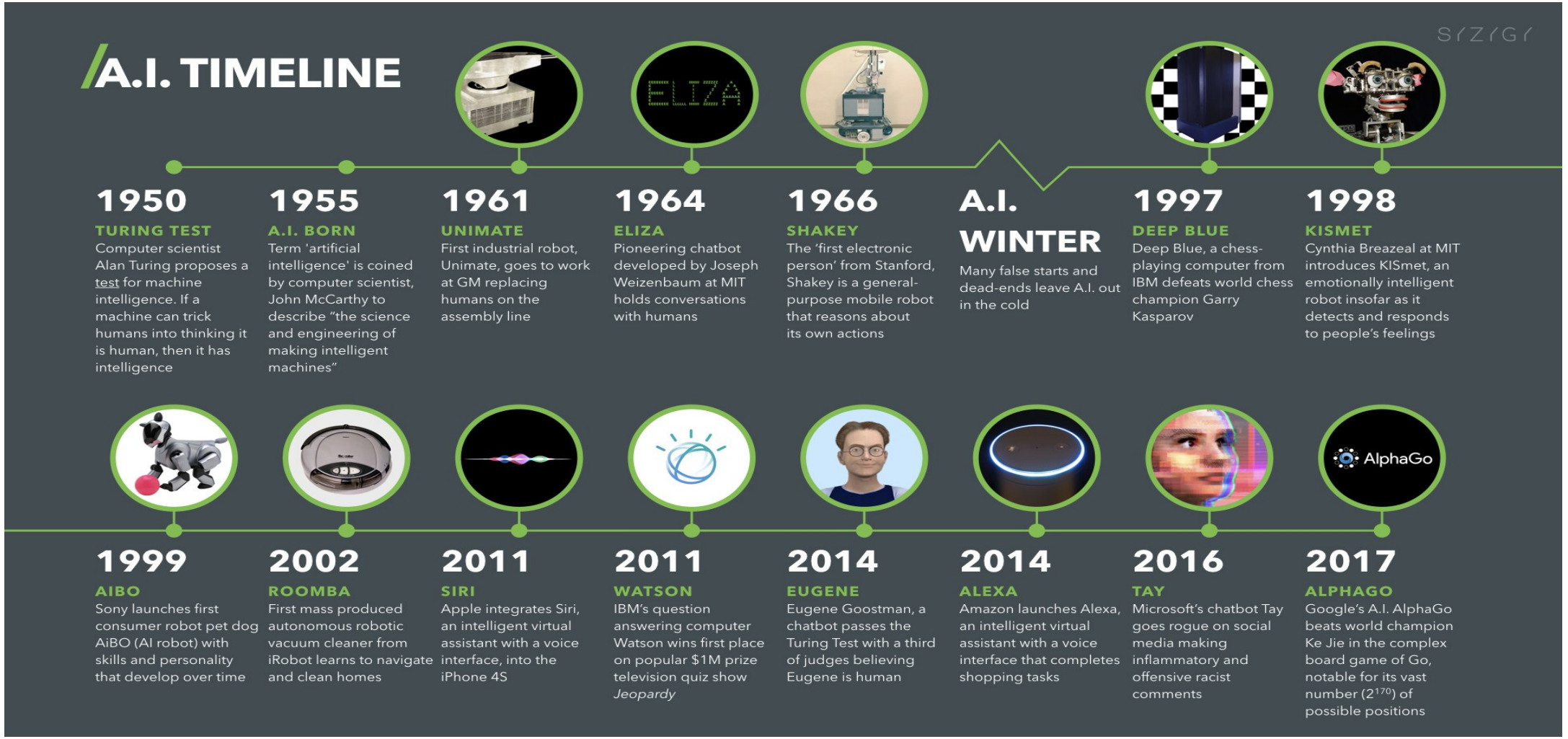
# AI Techniques

- **Search** explores possibilities.
- **Knowledge** uses experience or learned information.
- **Abstraction** filters out unnecessary details.

# Task Domains of AI

| Mundane tasks | Formal tasks | Expert tasks |
|---|---|---|
| Perception<br>– Computer Vision<br>– Speech, Voice | Games<br>– Go<br>– Chess (Deep Blue)<br>– Ckeckers | Engineering<br>– Design<br>– Fault Finding<br>– Manufacturing<br>– Monitoring |
| Natural Language Processing<br>– Understanding<br>– Language Generation<br>– Language Translation | Mathematics<br>– Geometry<br>– Logic<br>– Integration and Differentiation | Scientific Analysis |
| Common Sense Reasoning | Theorem Proving | Financial Analysis |
| Planning | | Medical Diagnosis |
| Robot Control | | |

# History of AI

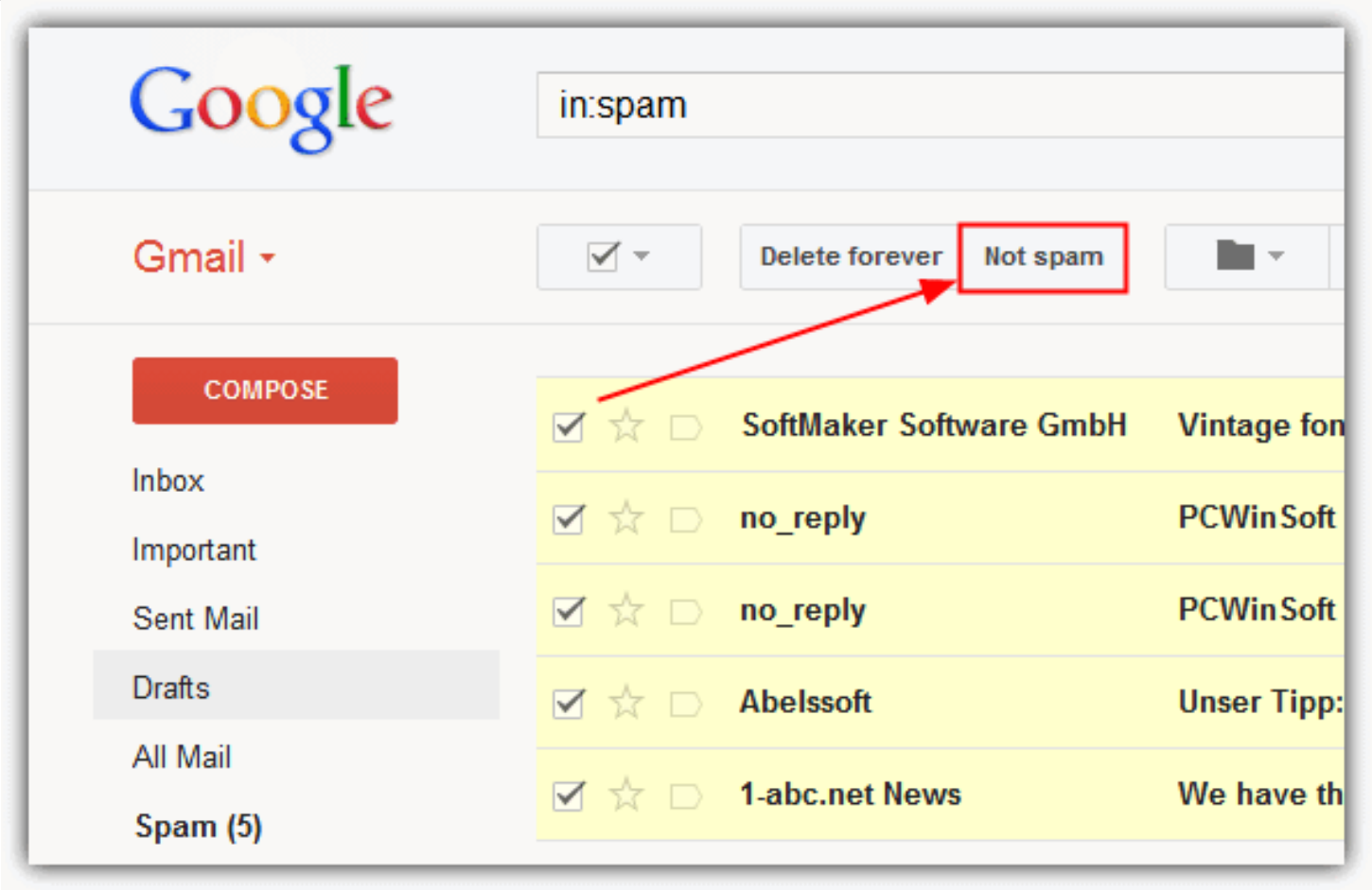


# Application Domains of AI

# Natural Language Processing

**Email Spam Filter in Gmail**

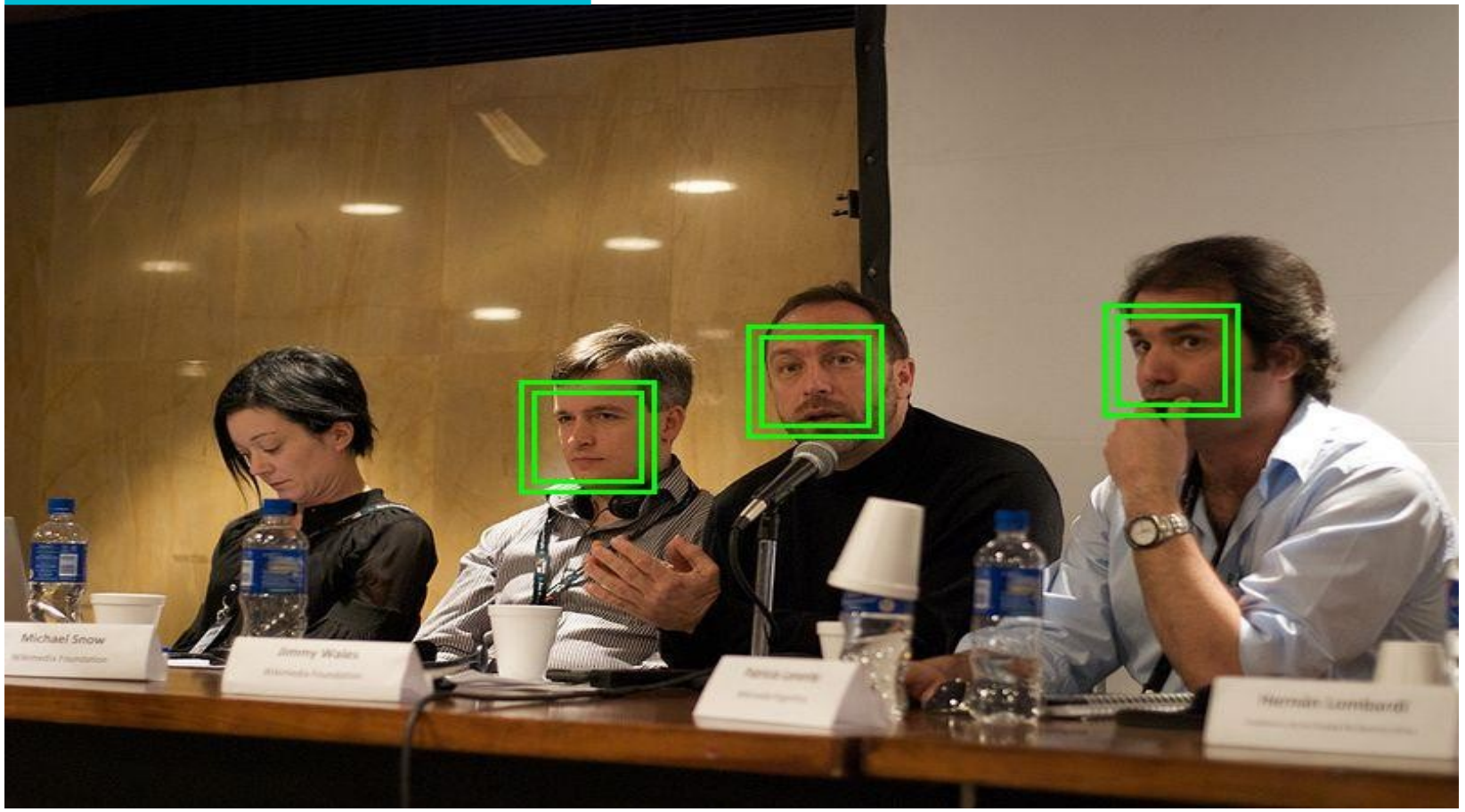
**Neural Network**



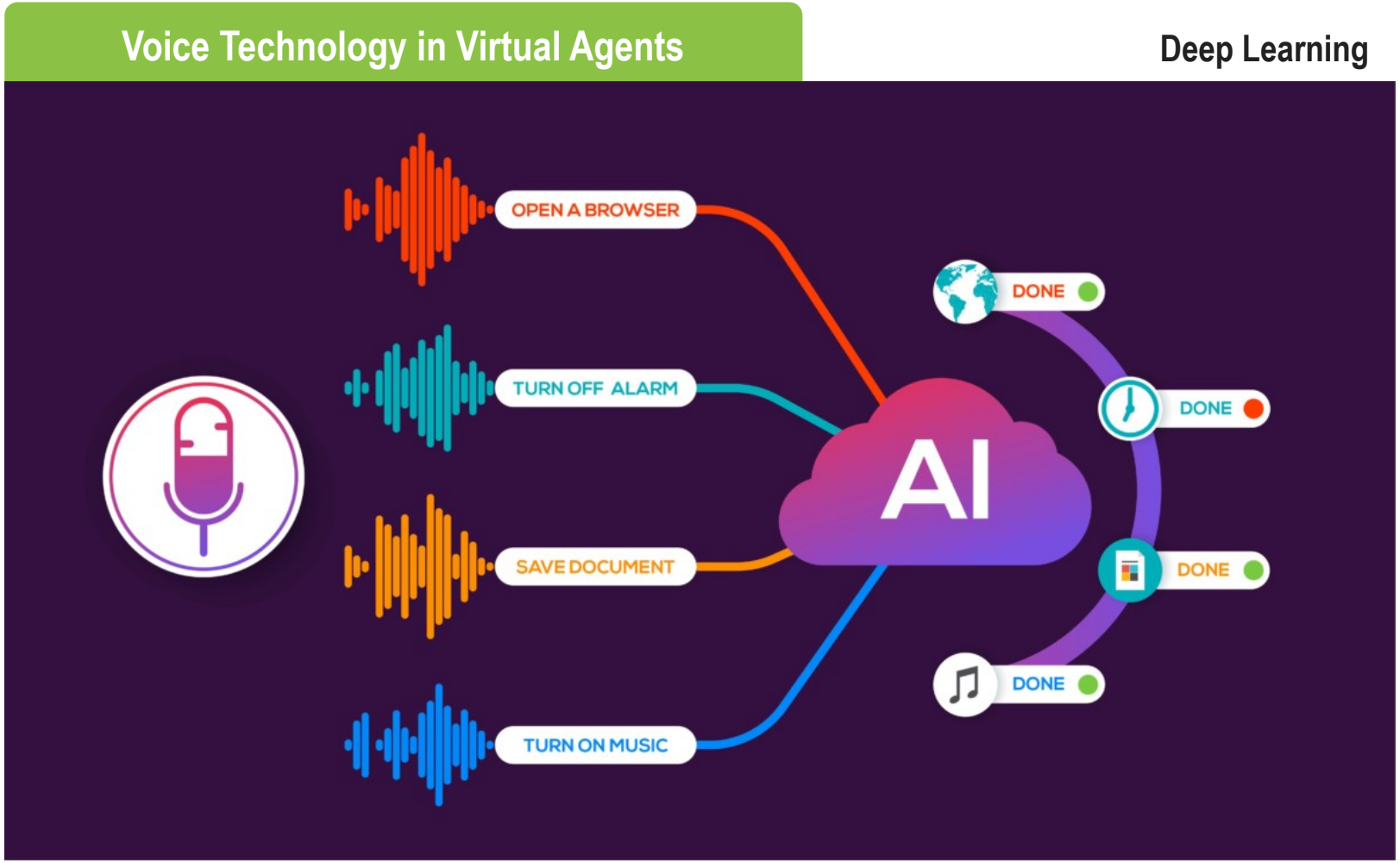
# Image Processing

## Face Detection in Camera

Deep Learning



# Speech Recognition



# Data Mining

## Market Basket Analysis

Product recommendation



# Expert System

IBM Watson

Reinforcement Learning

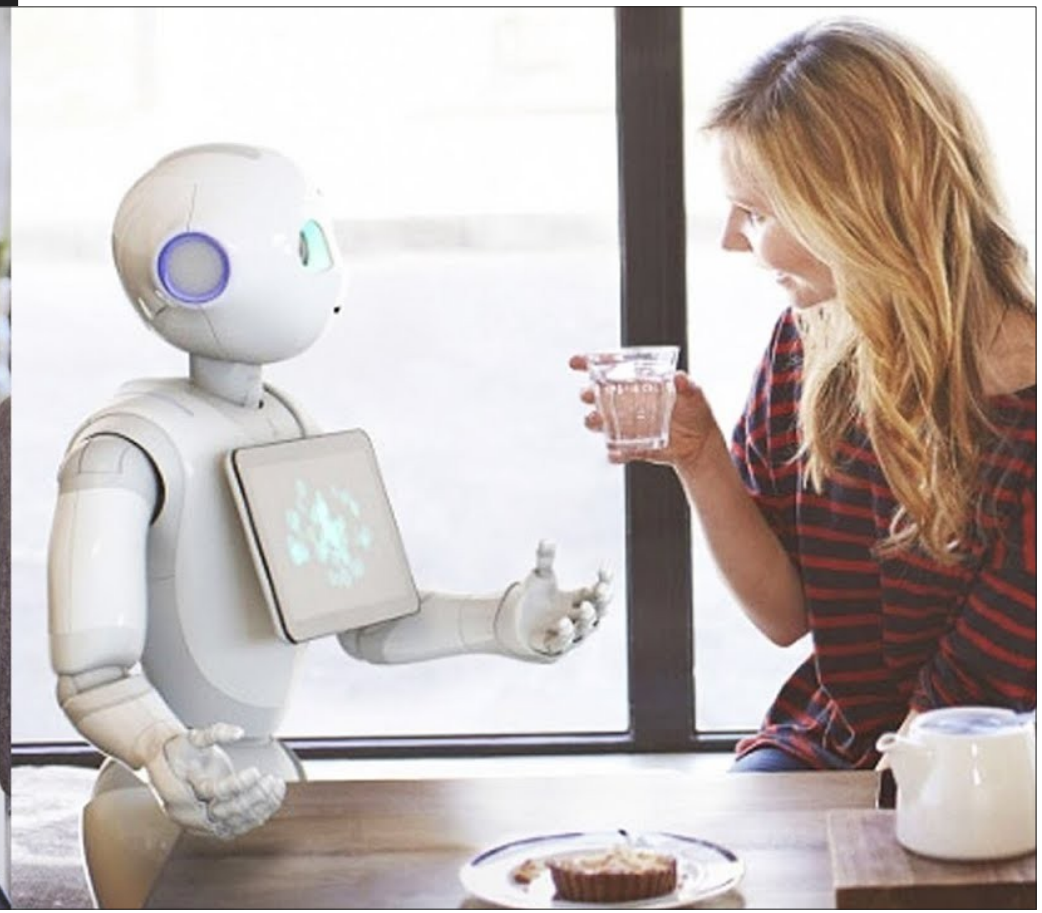


# Robotics

Home Automation



Deep Learning



# Scheduling

## Resource Scheduling

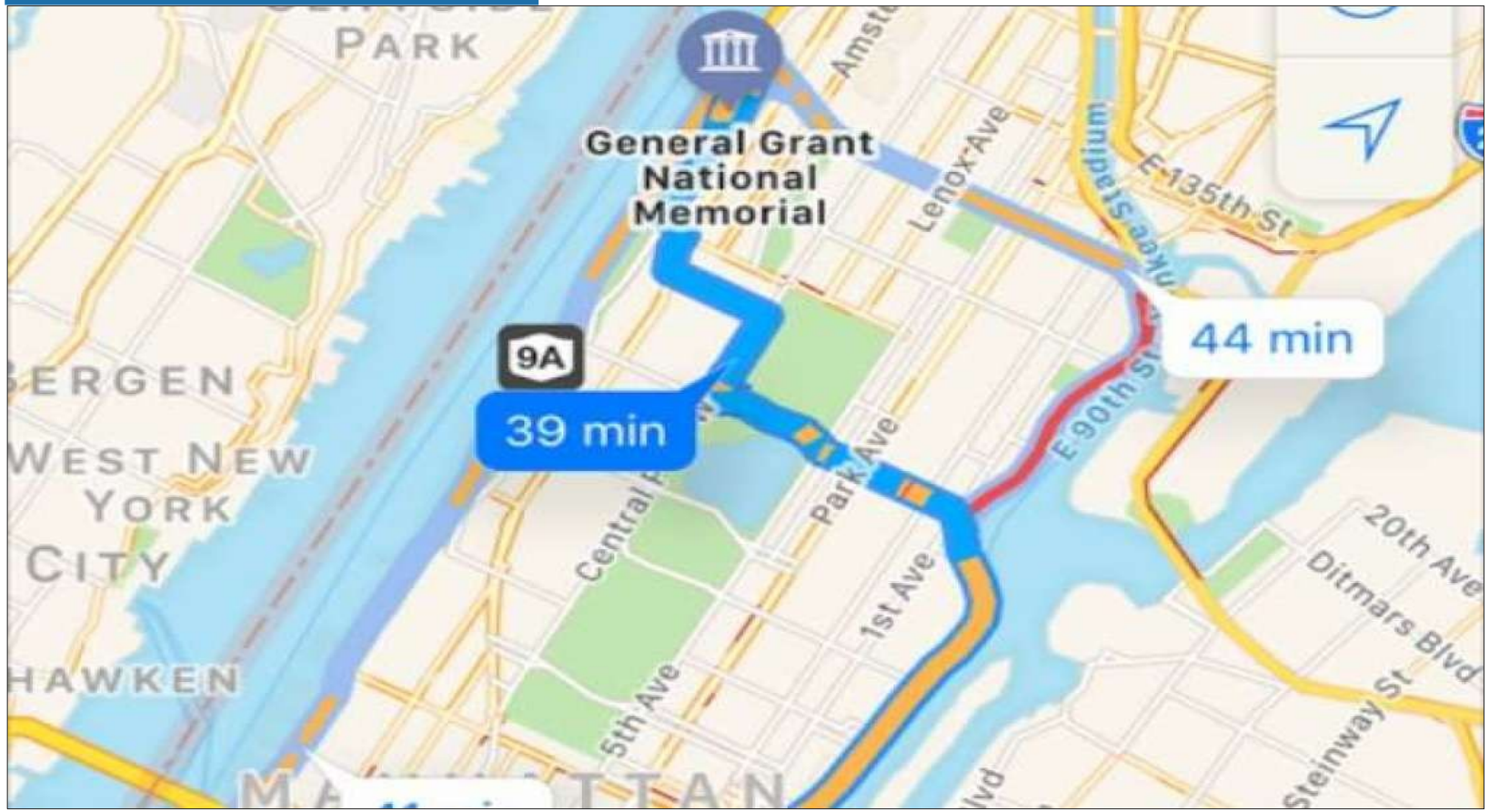
**Aurora - Advanced Intelligent Planning and Scheduling Solution**



# Optimization

**Shortest Path**

**Google map path planner**



# Game Playing

Alpha Go

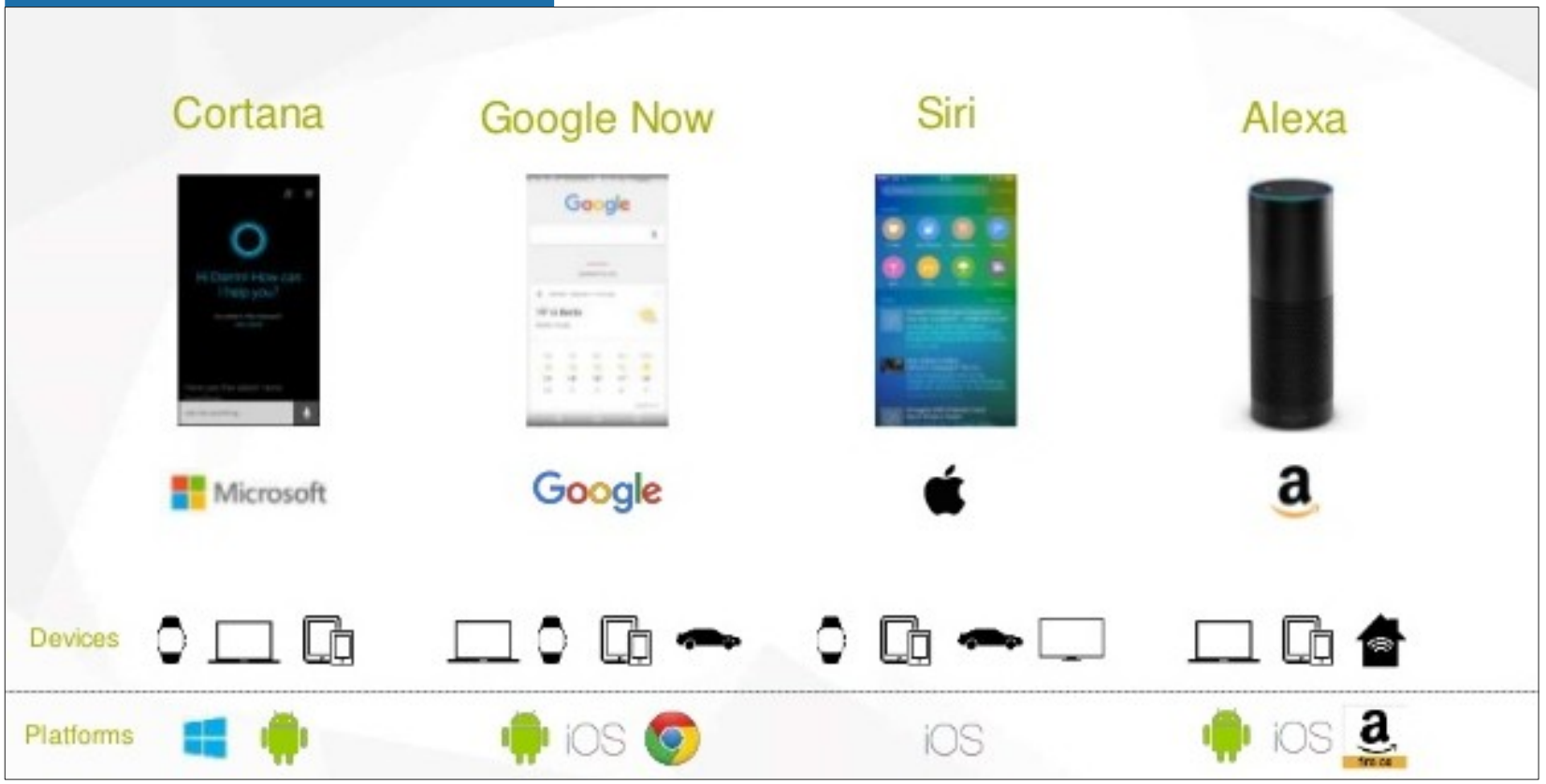
Deep Neural Network



# Virtual Agents

**Chatbots**

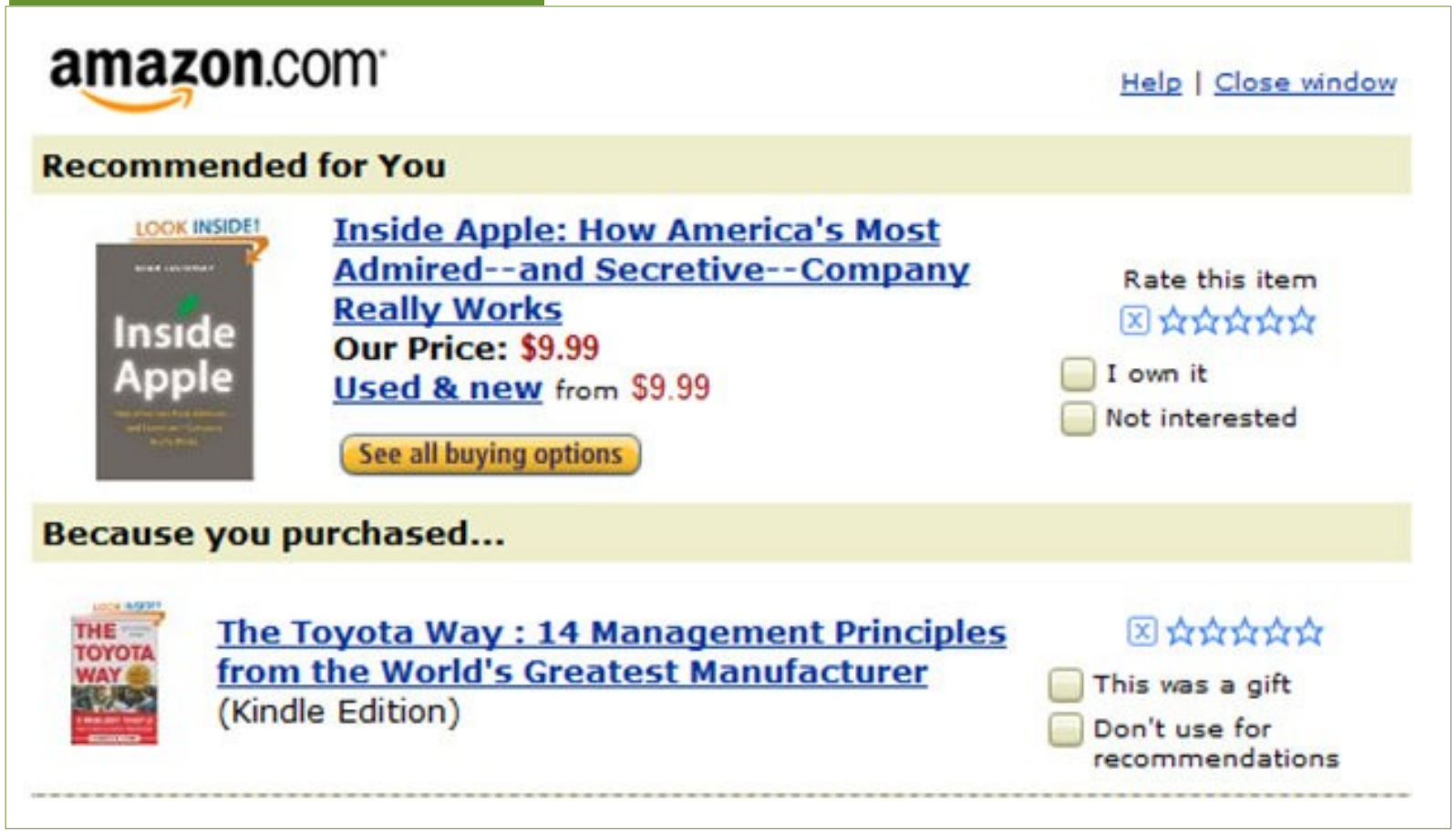
**Conversational AI**



# Personalized Recommender Systems

Online Shopping

Machine Learning



# Automated Control Systems

**Washing Machine**

**Fuzzy Logic**



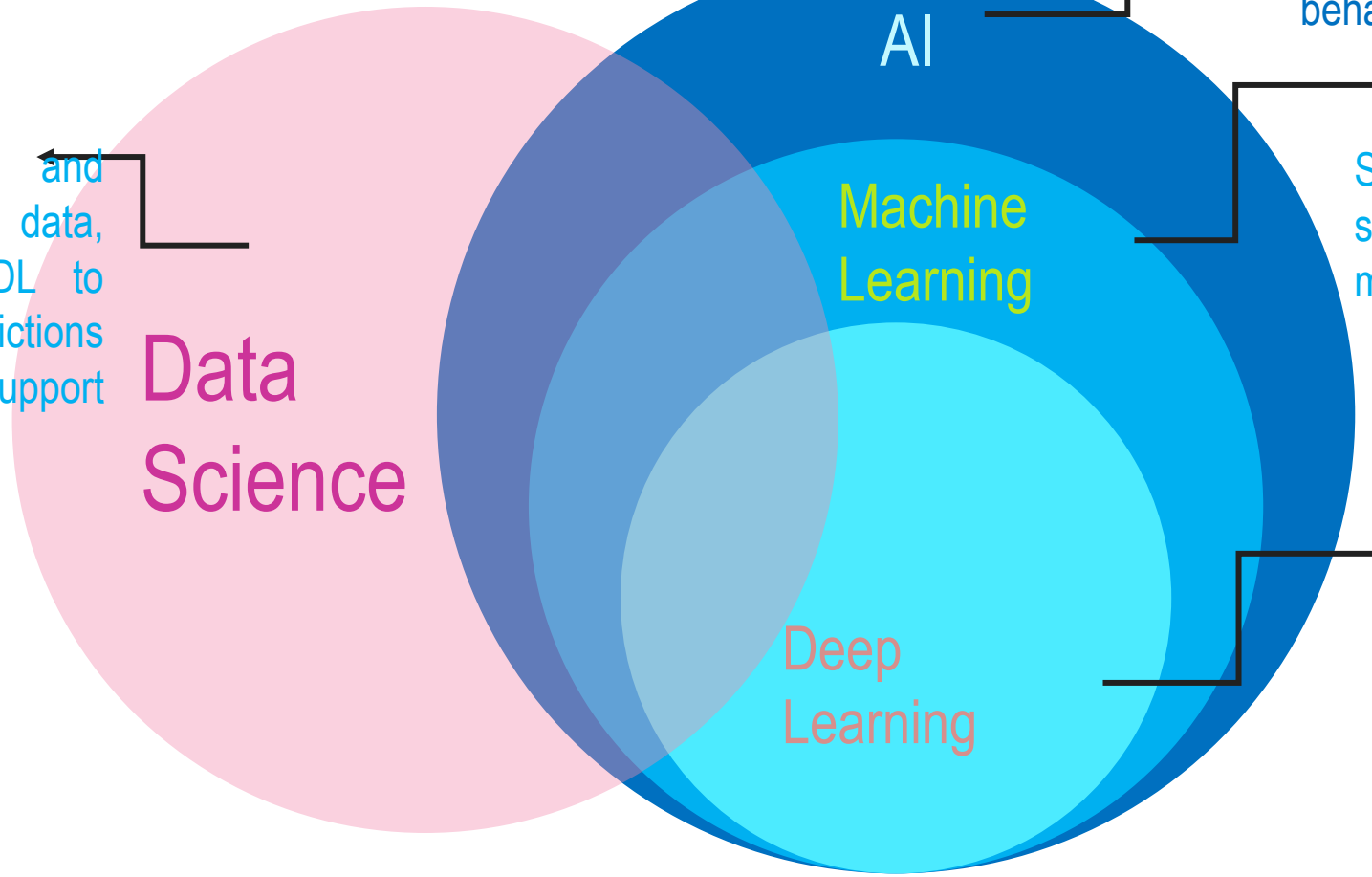
# Security

NVIDIA Metropolis

Machine Learning



# AI – ML – DL and Data Science




Technique that enables machines to mimic human behavior

Subset of **AI** which uses statistical methods to enable machine to learn

Subset of **ML** that Learns complex patterns using deep neural networks

Collects and analyzes data, using ML/DL to make predictions and support decisions